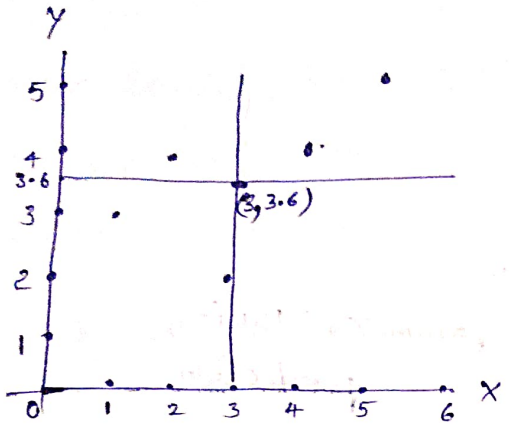
① Linear Regression



①

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})^2$
1	3	-2	-0.6	4
2	4	-1	0.4	1
3	2	0	-1.6	0
4	4	1	0.4	1
5	5	2	1.4	4
				$\Sigma = 10$

mean $\bar{x} = 3$, $\bar{y} = 3.6$

$(x-\bar{x})(y-\bar{y})$
1.2
-0.4
0
0.4
2.8

$\Sigma = 4$

Equation : $y = mx + c$

y = dependent variable

x = independent variable

m = slope

c = constant

②

$$m = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

$$= \frac{1.2 + (-0.4) + 0 + 0.4 + 2.8}{4+1+0+1+4}$$

$$= \frac{4}{10}$$

$$= 0.4$$

③

$y = 0.4 \times x + c$

$3.6 = 0.4 \times 3 + c$

$c = 2.4$

Now as per Calculations

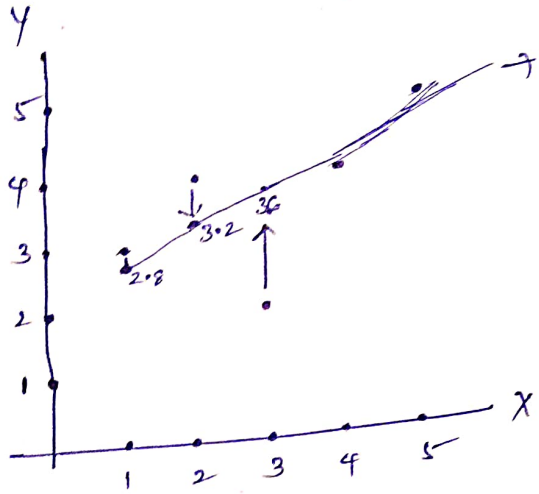
$m = 0.4$

$c = 2.4$

$y = 0.4x + 2.4.$

Here x holds 1, 2, 3, 4, 5.

			$\hat{y}$ = predicted	y act
$y =$	$0.4 \times 1 + 2.4$	$=$	2.8	3
	$0.4 \times 2 + 2.4$	$=$	3.2	4
	$0.4 \times 3 + 2.4$	$=$	3.6	2
	$0.4 \times 4 + 2.4$	$=$	4.0	4
	$0.4 \times 5 + 2.4$	$=$	4.4	5



→ This line is best fit line (Reg line)

$\hat{y} = mx + c$

Note: we can predict for x = any value.

if $x = 9$

$\hat{y} = 0.4 \times 9 + 2.4$

$= 6.$

final summary:

(1) Take x and y values

(2) Calculate $\bar{x}$ and $\bar{y}$ then $(x-\bar{x})$ and $(y-\bar{y})$ and $(x-\bar{x})^2$

and $(x-\bar{x})(y-\bar{y})$

(3) Calculate m and c

(4) apply for x.